Software Project Management
Assignment-11
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following are desirable characteristics of good software?

a. Huge size of code base
b. Correctness
c. Reliability
d. Usability
e. Hardware dependence

Correct Answer: b. Correctness
c. Reliability
d. Usability

Detailed Solution:

Desirable characteristics of good software include correctness (performs functions accurately), reliability (operates without failure under specified conditions), and usability (easy to learn and use). A huge code base (a) indicates complexity and higher maintenance costs, and hardware dependence (e) reduces portability—both are undesirable.

QUESTION 2:

Which statements regarding ISO 9001 are correct?

a. It mandates the use of a specific programming language.
b. It guarantees defect-free products.
c. It emphasizes management responsibility.
d. It requires a quality system to be established.
e. It includes process control requirements.

Correct Answer: c. It emphasizes management responsibility.
d. It requires a quality system to be established.
e. It includes process control requirements.

Detailed Solution:

ISO 9001 is a quality management system standard that focuses on processes, not specific technical outcomes. It emphasizes management responsibility (c), requires a documented quality system (d), and includes process control requirements (e). It does not mandate programming languages (a) or guarantee defect-free products (b).

QUESTION 3:

Which statements about ISO 9001 certification are correct?

a. Certification indicates that documented procedures are followed.
b. Certification automatically improves software reliability.
c. Certification alone does not guarantee product quality.
d. A certified organization may still have an ineffective process.
e. Certification removes the need for management oversight

Correct Answer: a. Certification indicates that documented procedures are followed.
c. Certification alone does not guarantee product quality.
d. A certified organization may still have an ineffective process

Detailed Solution:

ISO 9001 certification confirms that an organization follows documented procedures (a), but it does not guarantee product quality (c) or reliability (b)—it only ensures processes are defined and followed, not that they are optimal. A certified organization can still have ineffective processes (d), and certification does not remove the need for management oversight (e)—in fact, it emphasizes management responsibility.

QUESTION 4:

Which statements about the Initial (Level 1) maturity level are correct?

a. Processes are typically ad hoc and chaotic.
b. Processes are quantitatively managed.
c. Continuous process improvement is institutionalized.
d. Success depends heavily on individual effort.
e. It is the lowest maturity level.

Correct Answer: a. Processes are typically ad hoc and chaotic.
d. Success depends heavily on individual effort.
e. It is the lowest maturity level.

Detailed Solution:

At CMM Level 1 (Initial), processes are ad hoc, chaotic, and unpredictable (a), success depends heavily on individual effort rather than repeatable processes (d), and it is the lowest maturity level in the model (e). Quantitative management (b) is characteristic of Level 4, and continuous improvement (c) belongs to Level 5 (Optimizing).

QUESTION 5:

Which statements about the Defined (Level 3) maturity level are true?

a. Processes are primarily ad hoc.
b. Processes are documented and standardized.
c. A consistent process is followed across projects.
d. Organizational standards are established.
e. Continuous optimization is the primary focus.

Correct Answer: b. Processes are documented and standardized.
c. A consistent process is followed across projects.
d. Organizational standards are established.

Detailed Solution:

At CMM Level 3 (Defined), processes are documented, standardized, and consistently followed across projects, with organizational standards established (b, c, d). Ad hoc processes (a) describe Level 1 (Initial), and continuous optimization (e) is the focus of Level 5 (Optimizing).

QUESTION 6:

Which of the following correctly match a maturity level with its characteristic?

a. Repeatable (Level 2) – Disciplined process
b. Defined (Level 3) – Standard, consistent process
c. Initial (Level 1) – Continuously improving process
d. Managed (Level 4) – Predictable process
e. Optimizing (Level 5) – Ad hoc process

Correct Answer: a. Repeatable (Level 2) – Disciplined process
b. Defined (Level 3) – Standard, consistent process
d. Managed (Level 4) – Predictable process

Detailed Solution:

The CMM maturity levels are correctly matched as follows: Repeatable (Level 2) focuses on disciplined project management processes, Defined (Level 3) establishes standardized and consistent processes across the organization, and Managed (Level 4) uses quantitative metrics to achieve predictable process performance. In contrast, Initial (Level 1) is characterized by ad hoc, chaotic processes rather than continuous improvement (which belongs to Level 5 Optimizing), and Optimizing (Level 5) is about continuous process improvement, not ad hoc behavior.

QUESTION 7:

Which statements about CMM Level 5 are correct?

a. It is less mature than the Defined level.
b. It discourages process innovation.
c. It focuses on optimization and continuous improvement.
d. It includes Defect Prevention as a key process area.
e. It includes Technology Change Management as a key process area.

Correct Answer: c. It focuses on optimization and continuous improvement.
d. It includes Defect Prevention as a key process area.
e. It includes Technology Change Management as a key process area

Detailed Solution:

CMM Level 5 (Optimizing) is the highest maturity level, focusing on continuous process improvement, defect prevention, and technology change management to proactively enhance quality and efficiency. It is more mature than the Defined level (Level 3), not less, and it actively encourages process innovation rather than discouraging it.

QUESTION 8:

Which of the following are recognized Level 5 Key Process Areas (KPAs) in the SW-CMM?

a. Quantitative Process Management
b. Defect Prevention
c. Software Project Tracking and Oversight
d. Technology Change Management
e. Process Change Management

Correct Answer: b. Defect Prevention
d. Technology Change Management
e. Process Change Management

Detailed Solution:

In the SW-CMM, Level 5 (Optimizing) focuses on continuous process improvement through three Key Process Areas: Defect Prevention (to proactively eliminate errors), Technology Change Management (to adopt beneficial new technologies), and Process Change Management (to drive ongoing process enhancements). Quantitative Process Management belongs to Level 4 (Managed), and Software Project Tracking and Oversight is part of Level 2 (Repeatable), making (b), (d), and (e) the correct Level 5 KPAs.

QUESTION 9:

Which of the following statements correctly describe Quality Assurance (QA)?

a. QA focuses on improving development processes.
b. QA is limited to software testing.
c. QA begins only after coding is completed.
d. QA aims to prevent defects from occurring.
e. QA is process-oriented.

Correct Answer: a. QA focuses on improving development processes
d. QA aims to prevent defects from occurring.
e. QA is process-oriented

Detailed Solution:

Quality Assurance (QA) is a process-oriented approach that focuses on preventing defects by improving development and management processes throughout the lifecycle, starting from project inception. It is not limited to testing (which is Quality Control) and does not begin only after coding—it spans the entire development process.

QUESTION 10:

Which of the following statements correctly describe Quality Control (QC)?

a. QC aims primarily at process improvement.
b. QC is performed only during requirements analysis.
c. QC focuses on identifying defects in software products.
d. QC is product-oriented.
e. QC includes activities such as testing and inspections.

Correct Answer: c. QC focuses on identifying defects in software products.
d. QC is product-oriented.
e. QC includes activities such as testing and inspections.

Detailed Solution:

Quality Control (QC) is a product-oriented process that focuses on identifying defects in the actual work products through activities like testing, inspections, and reviews. It is performed throughout the development lifecycle—not just during requirements analysis—and its goal is defect detection, not process improvement (which is the aim of QA).

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